Definition (Strict monoidal stacking semicategory)
A *strict monoidal stacking semicategory* is a stacking semicategory $\langle \mathbf{C}, \otimes, \otimes \rangle$ with

<u>Constituents</u>

1. an object $\mathbf{1} \in \mathrm{Ob}_{\mathbf{C}}$, called the *monoidal unit*

<u>Conditions</u>

1. For any object X of $\mathbf{C}$,

$$X \otimes \mathbf{1} = X \qquad \text{and} \qquad \mathbf{1} \otimes X = X.$$

2. The monoidal unit $\mathbf{1}$ has an identity morphism $\mathrm{id}_{\mathbf{1}}$, and for any morphism $f : X \to Y$,

$$f \otimes \mathrm{id}_{\mathbf{1}} = f \qquad \text{and} \qquad \mathrm{id}_{\mathbf{1}} \otimes f = f.$$